

James Bai

Engineer @ Xero · 5+ yrs · .NET · AWS · React · LLM firefighter

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Wellington · NZ

```
/**
 * Software engineer with 5+ years shipping .NET and AWS systems at Xero, now leading AI/MCP
 * and Microsoft Office data-flow work after App Store billing systems.
 *
 * I use AI heavily, but my value is knowing where to point it, how to verify the result, and how
 * to repair the system when it gets things wrong. Claude Code and Codex are daily tools.
 * My side project is a ~457,000 line AI powered ERP I built and operate alone, with ~10 daily users.
 *
 * @objective Senior software engineering roles
 */
```

experience

Xero · Engineer

2023.11 – present

stack: C# / .NET 8/10 · Node.js · Python · AWS Lambda · SQS · SNS · DynamoDB · AWS Glue · React / TypeScript · Postgres · MCP

CURRENT — AI / MCP / RAG

- ai/current** Selected to the team owning Xero's official **AI/MCP surface**, including the **Xero MCP server**. Day-to-day work covers AI platform integration, **LLM tool use**, agent workflows, retrieval/context design, eval checks, observability, privacy, and safety guardrails.
- office-flow** **Feature lead** on Microsoft Office support for **Word, Excel, and PowerPoint**, letting Xero data flow straight into Office files as structured content so users no longer copy/reformat numbers from Xero or build charts manually.

PREVIOUS — billing / finance systems

- billing-core** **Feature lead** on major parts of the global **App Store usage-based billing** rewrite. Designed and shipped the **.NET / C# APIs** and Postgres schema that became the **billing core**. Also built the **AWS Glue** aggregation pipeline. **Went live with one minor post-launch bug.**
- merchant-spa** Co-designed and shipped **React / Refine / MUI SPA** backed by a **.NET BFF** for App Store merchant self-service.
- invoice-recovery** **Feature lead** on the manual-invoicing flow that **recovered hundreds of thousands of dollars** in overdue invoices. Leadership shout-out.
- us-tax-support** On-call owner for the **US Sales Tax** service supporting Xero's US merchants: production incidents, schema fixes, customer-facing data corrections.
- also** Deep Embed Service · CodeRed CI/CD · shared frontend libs · PCI compliance · API latency · associate mentoring.

Section6 · Software Consultant

2021.03 – 2023.11

stack: C# / .NET · Azure · React / TypeScript · Python · Kubernetes

- virtual-cfo** Built **.NET / C# APIs** on **Azure Functions** integrating Xero & BambooHR, paired with **React / TypeScript / Fluent UI** dashboards for finance teams. APIs processed **66k+ records/year** in seconds.
- bnz-migration** Red Hat partnership. Python/Bash automation migrated **270 of 350** Java apps off legacy Kubernetes within a year.
- vodafone-dx** Owned **Payment, SMS, and Email gateways** on **.NET / C# · Azure · Kubernetes**, serving Vodafone NZ's digital experience platform.
- also** Multi-client consulting across banking, telco, and SaaS · CI/CD on TeamCity & GitHub Actions · code review and junior mentoring.

side-projects

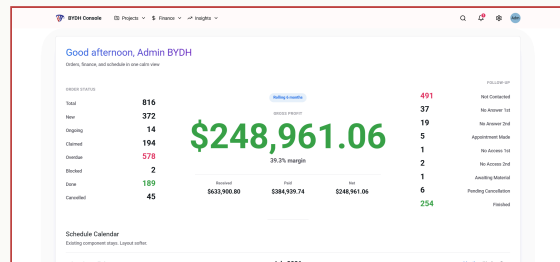
BYDH ERP • AI powered, solo build

react • ts • .net • aws • postgres
case study at /bydh

~457,000 LOC • ~10 daily users • 1 maintainer

A custom ERP for a construction business. **Designed, built, and operated end to end** as an AI native system across web and native mobile.

- bill-parser** **Gemini** parses inbound bills, classifies them, and drafts payable/receivable records with review before save.
- ai-quoting** **MCP** lets **ChatGPT** drive quoting via tool calls, context, pricing lookup, preview, and controlled save.
- insights** Cashflow, P&L, and project profitability dashboards.
- mobile** **React Native + Expo** for site use, sharing auth and **AWS Cognito** with web.



live-products

- Success Health** **Built and operate** clinic booking and customer-facing flows for a Wellington physio.
successhealth.co.nz
- AHU Handyman** Public site for an Australian handyman business, focused on service discovery, lead capture, and simple production deployment.
ahuhandyman.com.au

ml-vision

- pixel-style-transfer** Neural style transfer in **PyTorch**: Gatys + AdaIN + Johnson, VGG-19 features, pixel-art grid-preservation loss, ControlNet baseline.
- cv-self-training-counter** CV auto-label (Dual-Otsu + SimpleBlobDetector) → Label Studio → **YOLOv8s** self-training loop → CoreML export.
- receipt-ocr** **Florence-2** VLM + **PaddleOCR** with line-grouping algorithm and AWS Lambda container deploy plan.

skills

- lang** C# / .NET • TypeScript • Python
- web** React • Redux • Next • MUI • Fluent • Next UI
- mobile** React Native • Expo
- data** Postgres • MySQL • DynamoDB • Dapper • EF • Flyway
- msg** SQS • SNS
- cloud** AWS (Lambda, SQS, SNS, DynamoDB, Glue, ECS, S3, RDS, **Cognito**, CloudFront) • Azure
- platform** K8s • Docker • Terraform • CloudFormation
- ci/cd** GitHub Actions • TeamCity • Jenkins
- ai** Claude Code • Codex • Gemini • MCP • LLM tool use • agents • RAG/context • evals

education + cert

PGDipSci (Merit), Computer Science

University of Auckland • 2020 — 2024

AWS Certified Developer Associate

Amazon Web Services • 2026

Azure Developer Associate • Microsoft

2020 — 2022

BSc, CS & Statistics (AI specialisation)

Victoria Univ. of Wellington • 2015 — 2018

RHCSA • Red Hat

2022 — present